CS & IT ENGINEERING

DIGITAL LOGIC

Combinational circuit



Lecture No. 1



By - CHANDAN SIR

Recap of Previous Lecture



K-MAP

Topics to be Covered



Combinational circuit

#Q. Let a function F which has 3 input variables (x,y,z). The function F will be high only when atleast two of the input variables are set to high. Draw the K-Map for the given function. Let the number of PI in K-Map = 'a' and the number of EPI in K-Map = 'b'. Find the quadratic mean of 'a' and 'b' _____

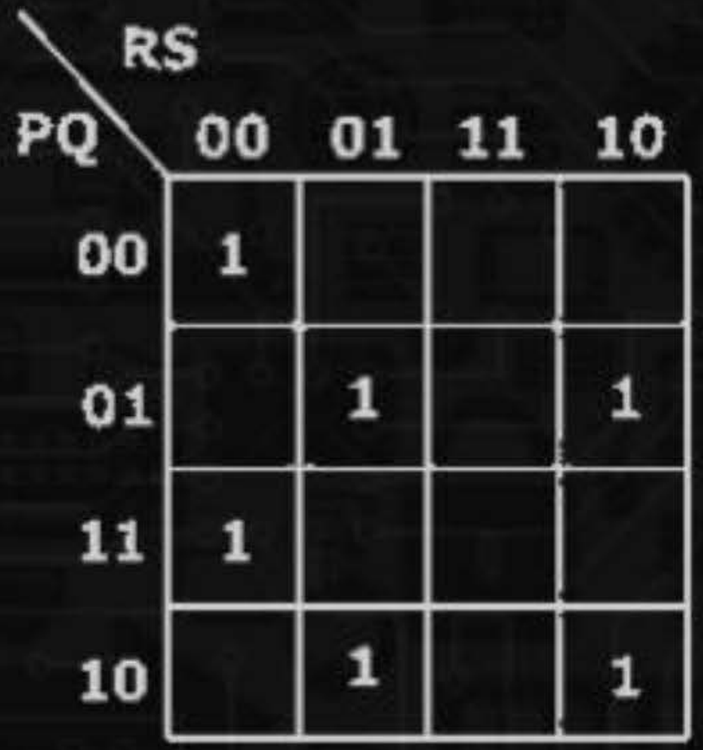
A. 3

B. 4

C. 5

D. 6

#Q. Find the number of Prime implicants & Essential prime implicants in the given K



A. 4, 5

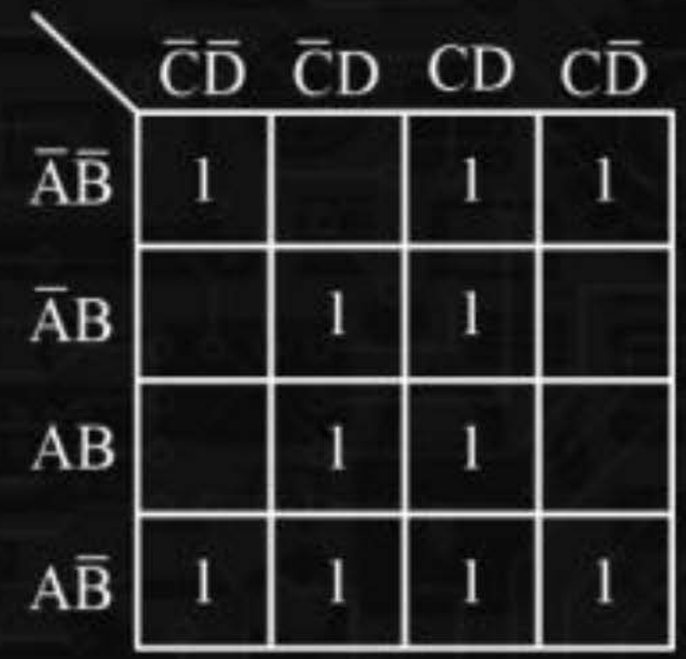
B. 0, 6

C. 6, 0

D. 6, 6

(MCQ)

#Q. Consider the K-map given below:
The EPI term will be:



	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1		1	1
$\bar{A}B$		1	1	
AB		1	1	
$A\bar{B}$	1	1	1	1

A $BD, A\bar{B}$

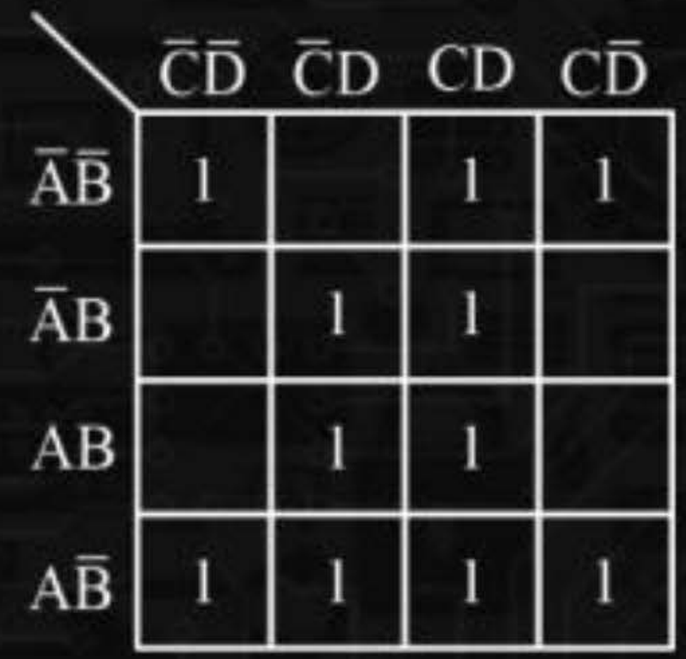
B $A\bar{B}, CD$

C $BD, \bar{B}\bar{D}$

D $BD, \bar{B}C$

(MCQ)

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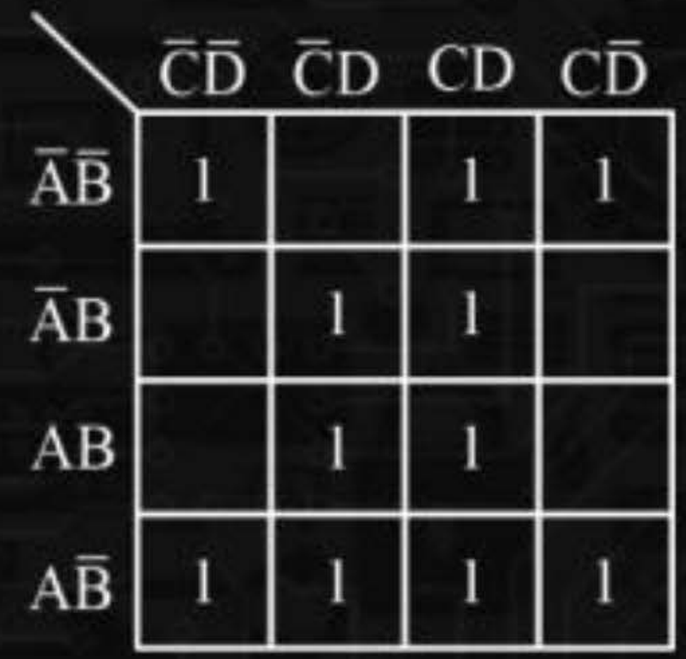
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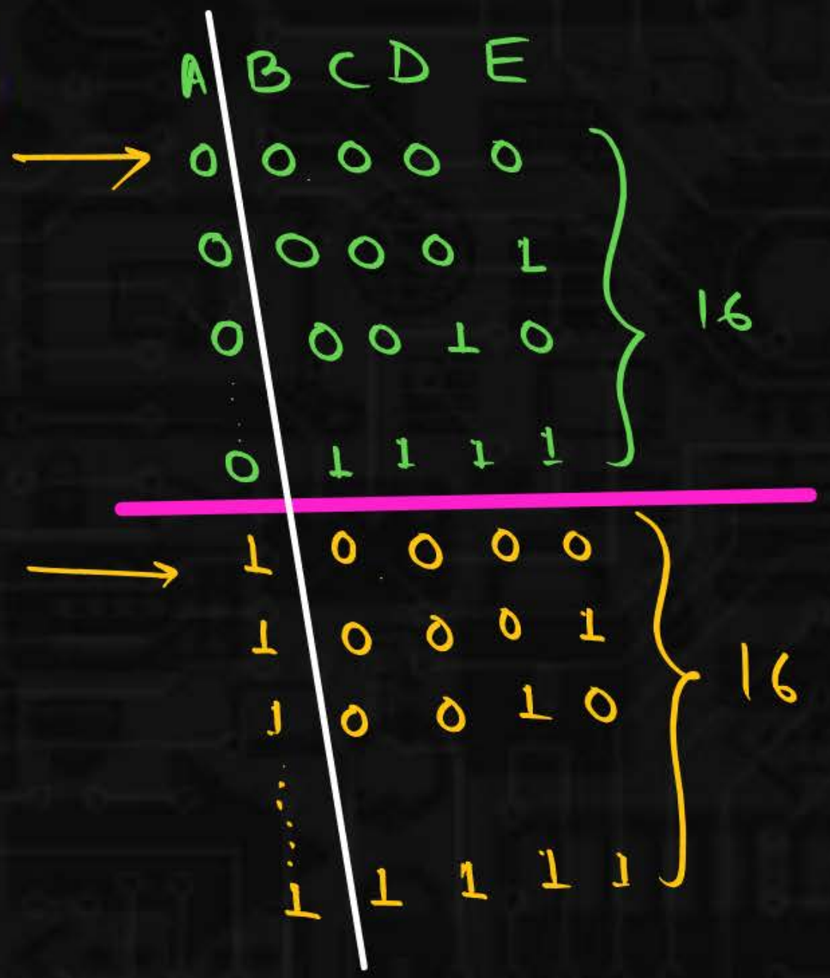
A $BD, A\bar{B}$

B $A\bar{B}, CD$

C $BD, \bar{B}\bar{D}$

D $BD, \bar{B}C$

5 Variable k-map



PRIME IMPLICANTS AND ESSNTIAL PRIME IMPLICANTS

5 Variable k-map

$\bar{A}$

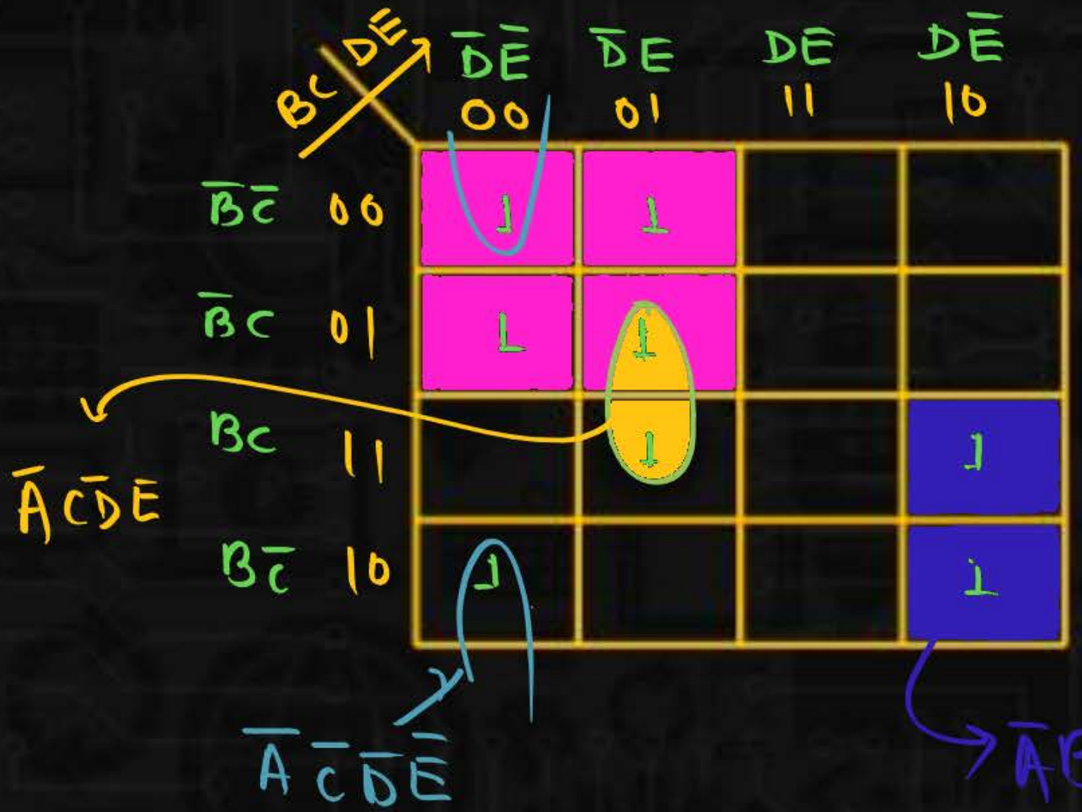
BC \ DE	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

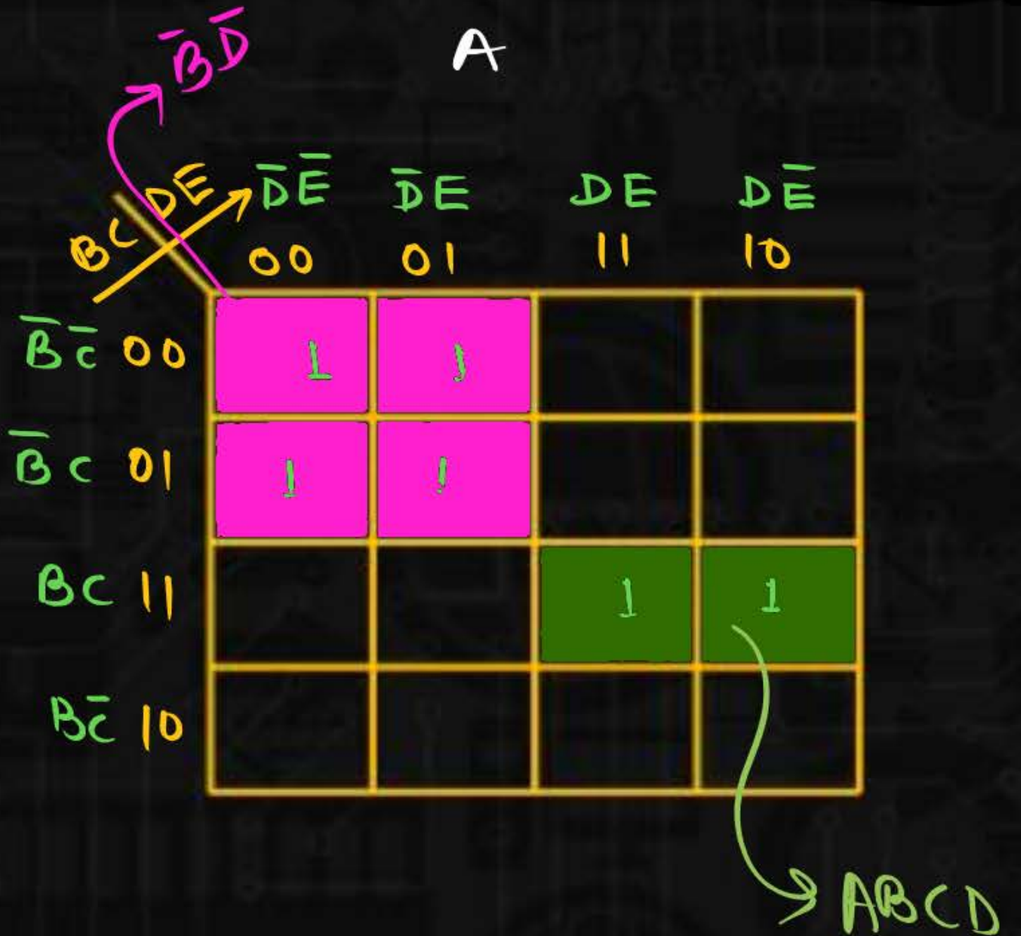
A

BC \ DE	00	01	11	10
00	16	17	19	18
01	20	21	23	22
11	28	29	31	30
10	24	25	27	26

#Q. $f(A, B, C, D, E) = \sum m(0, 1, 4, 5, 8, 10, 13, 14, 16, 17, 20, 21, 30, 31)$

$\bar{A}$





$\bar{B}\bar{D} + ABCD + \bar{A}C\bar{D}E + \bar{A}\bar{C}\bar{D}\bar{E} + \bar{A}B\bar{D}\bar{E}$

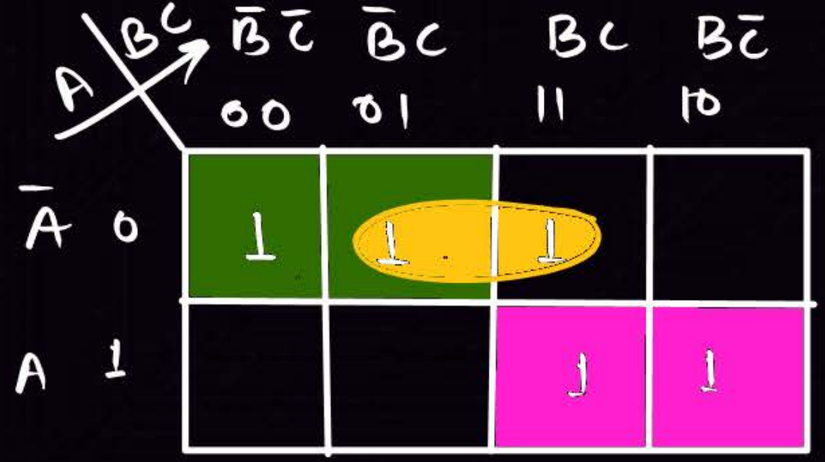
Hw

$\overline{A}$

BC \ DE	00	01	11	10
00	1		1	1
01		1	1	
11		1	1	
10	1		1	1

A

BC \ DE	00	01	11	10
00	1			1
01	1	1	1	
11		1	1	
10	1		1	1



EPI

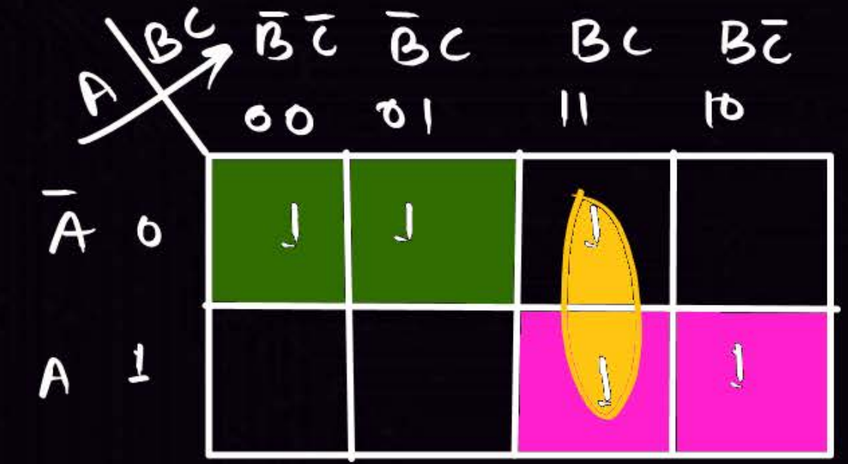
$= \bar{A}\bar{B} + AB + \bar{A}C$ ✓

$PI = \{\bar{A}\bar{B}, AB, \bar{A}C, BC\}$

$I = 5$

$PI = 4$

$EPI = 2$

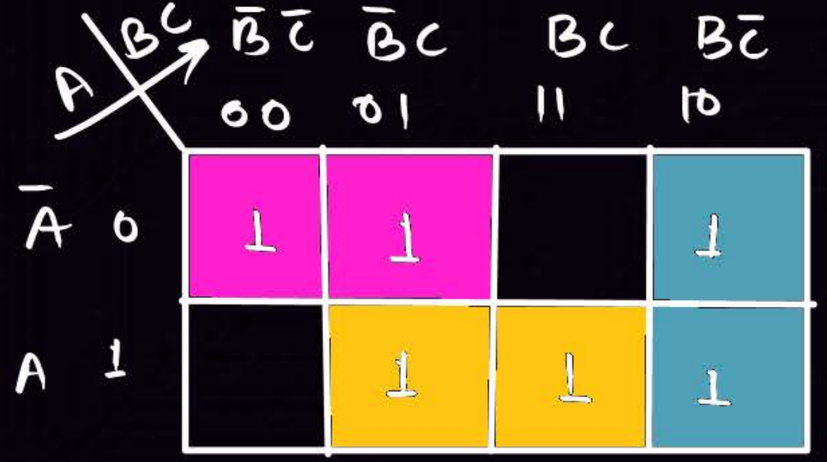


$= \bar{A}\bar{B} + AB + BC$ ✓

SPI

(SPI) Selective PI = 1

(RPI) Reduced PI = 1



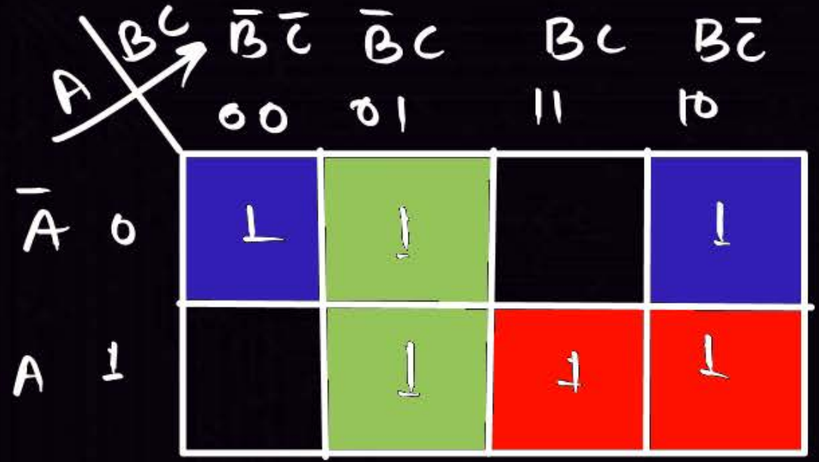
$= \bar{A}\bar{B} + AC + B\bar{C}$

$PI = \{\bar{A}\bar{B}, AC, B\bar{C}, \underline{\bar{A}\bar{C}, \bar{B}C, AB}\}$

$I = 6$

$PI = 6$

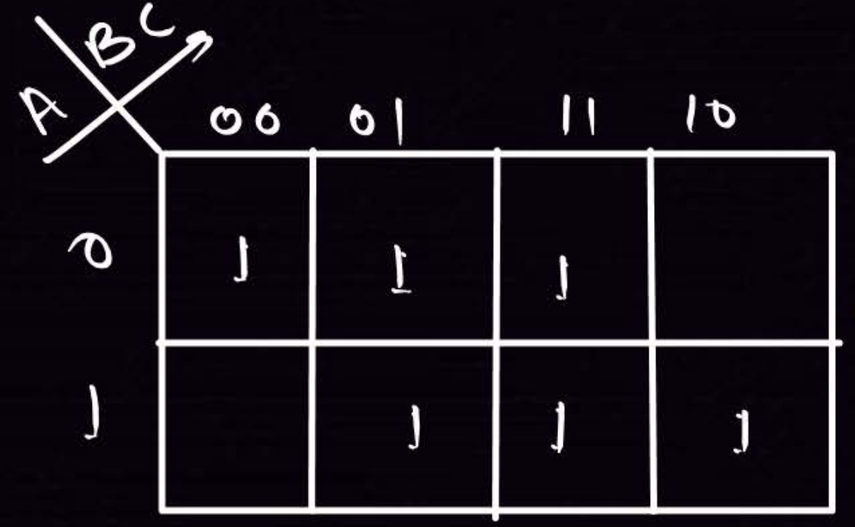
$EPI = 0$



$= \underline{\bar{A}\bar{C} + \bar{B}C + AB}$ ✓

(SPI) Selective PI = 3

(RPI) Reduced PI = 3



$= \bar{A}\bar{B} + AB + C$

$PI = \{\bar{A}\bar{B}, AB, C\}$

$I = 6$

$PI = 3$

$EPI = 3$

$SPI = 0$

$RPI = 0$

PI = { $\bar{A}\bar{C}D$, $\bar{A}BC$, $AB\bar{C}$, ACD, BD }

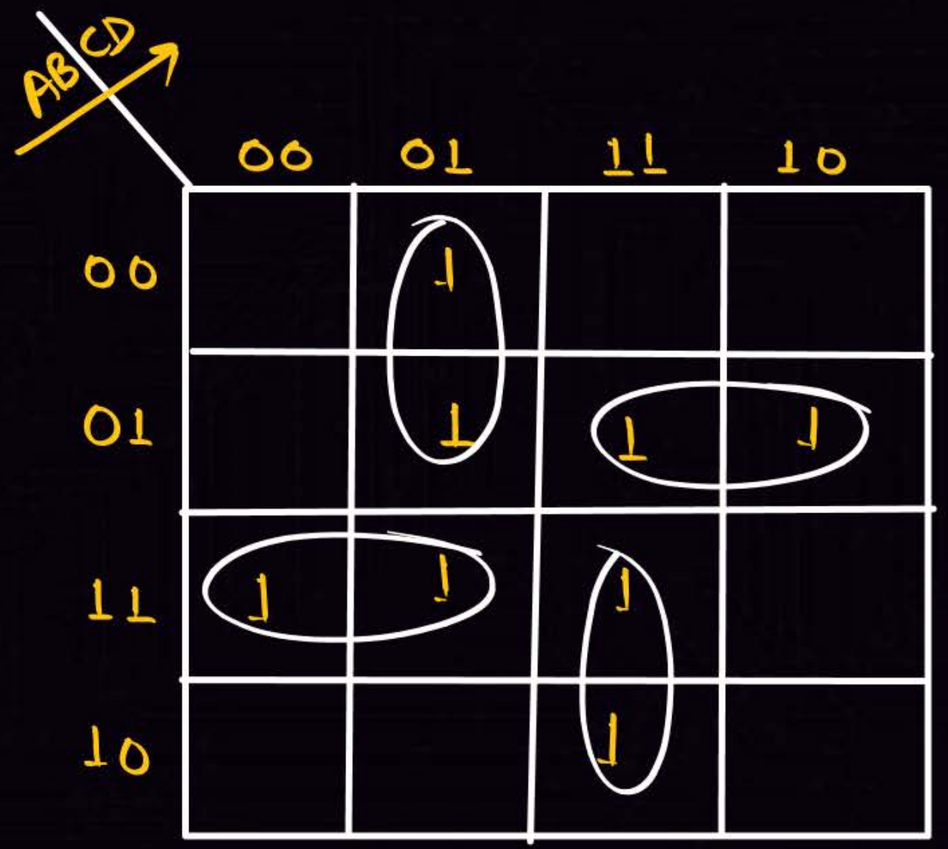
I = 8

PI = 5

EPI = 4 ✓

SPI = 0

RPI = 1



$= \bar{A}\bar{C}D + \bar{A}BC + AB\bar{C} + ACD$

COMBINATIONAL CIRCUIT :→

↳ Circuit without Feedback or memory

↳ Static Circuit

Designing :-

Step 1 :→ Find the number of inputs & outputs.

Step2 → Write the Truth table.

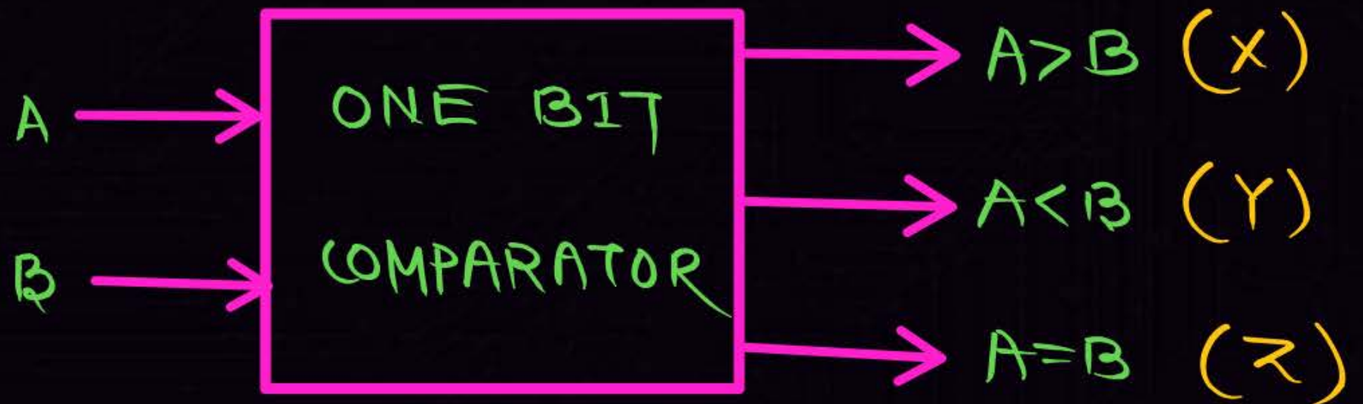
Step3 → Write the Logical Expression

Step4 → Minimization

Step5 → Hardware Implementation.

Q Design a One bit comparator?

Step 1:



Step 2.

			(A>B) X	(A<B) Y	(A=B) Z
⇒	0	0	0	0	1
	0	1	0	1	0
	1	0	1	0	0
⇒	1	1	0	0	1

Step 3: $X(A>B) = A\bar{B}$

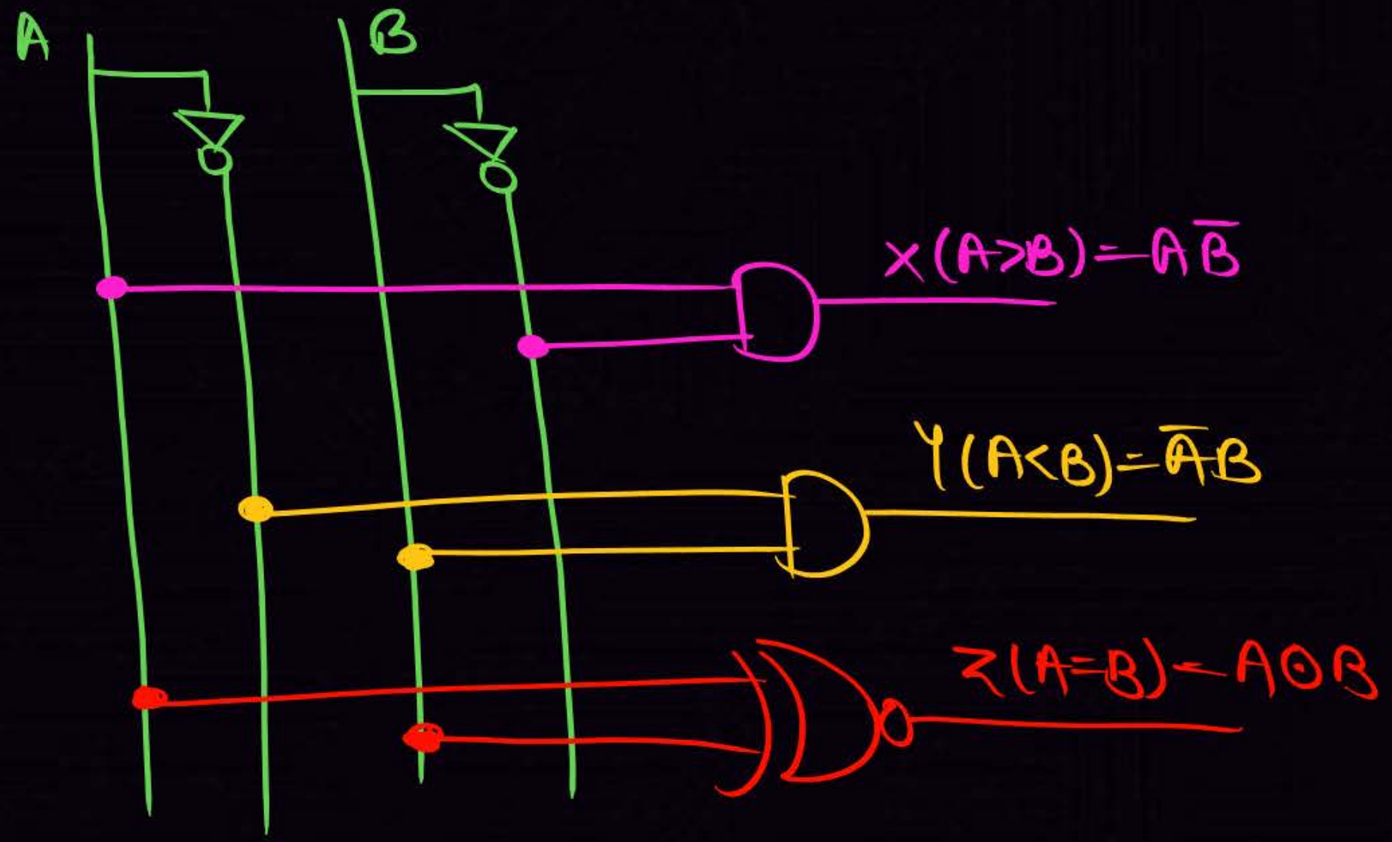
$Y(A<B) = \bar{A}B$

$Z(A=B) = \bar{A}\bar{B} + AB = A \odot B$

Step 4: Minimization

→ already minimize

Step 5:





2 Minute Summary

Combinational circuit

Thank you

GW

Soldiers !

